Product Requirements Document (PRD)

Dynamic Trust Infrastructure — Hackathon MVP

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1. Overview

A Trust API and platform-independent system that sits on top of existing KYC providers and delivers configurable trust signals, personal digital guarantees, portable work credentials, and (for direct jobs) escrow-backed protection, so platforms and individuals can safely engage informal and gig workers.

Problem
Commodity KYC only answers “Who is this person?” It does not answer “How much can I trust this person for this job?”

Result:

- Platforms suffer high cutout (users go offline after matching).
- Households and individuals face elevated risk with no reliable recourse.
- Workers cannot carry reputation across platforms.

Solution

- Leverage existing KYC providers for identity.
- Provide a configurable Trust Scoring Engine that returns a boolean decision + confidence score.
- Enable personal guarantor instruments with electronic signature.
- Allow workers to import/export verified work credentials.
- Offer escrow + micro-insurance **only for platform-independent jobs**.
- Normalise differently structured data from multiple platform types.

Beachhead

In-home domestic services, with architecture that supports other verticals (handyman, ride-hailing, etc.).

Timeline

1-month hackathon MVP.

2. Major Use Cases

Use Case 1 — Platform Verification API (B2B)

Gig, domestic, and related platforms integrate the API for onboarding validation, trust queries, outcome recording, and guarantor management.

Use Case 2 — Portable Worker Credentials

Workers import and export verified work history and trust records across platforms so they do not start from zero.

Use Case 3 — Platform-Independent Jobs

Individuals and workers use the system directly for jobs outside major marketplaces. This mode includes personal guarantor flow + escrow + micro-insurance + anti-gamification reviews.

3. Goals & Success Metrics

Must demonstrate

- API returns boolean + confidence for onboarding/validation.
- Platforms can configure baselines and weights by platform type.
- Data normalisation layer accepts varied platform payloads.
- Personal guarantor flow with electronic signature works end-to-end.
- Workers can import/export credentials.
- Platform-independent jobs support escrow (Vesti) + micro-insurance.
- Job completion confirmation feeds the scoring engine.
- Side-by-side On-Platform vs Off-Platform experience.
- Basic anti-gamification rating for independent jobs.

Explicit Non-Goals

- Building our own KYC stack
- Escrow on platform-integrated jobs
- Live insurance provider integrations (stubs only)
- Production DID/VC infrastructure (practical credential sharing is sufficient)
- Full multi-vertical production data pipelines
- Native mobile apps

4. Core Features (Prioritised)

P0 – Must Ship

1. Trust Scoring Engine

- Inputs: work history signals (completion rate, disputes, no-shows, repeat-hires, ratings, cancellations, quality flags, etc.).
- Output:
    - decision: true/false (meets platform-configured baseline)
    - confidence: numeric score
    - Band: Low / Medium / High / “Building History” (for workers with insufficient data)
- Platforms can fine-tune thresholds, weights, and required signals according to platform type.

2. Data Normalisation Layer

- Accepts differently structured payloads from different platform types.
- Maps them into a common internal signal format so the scoring engine remains consistent.

3. Work Outcome Ingestion

- Platforms (and independent flow) post structured outcome events.
- Events update the worker’s trust record and portable credentials.

4. Personal Guarantor Flow (both API and independent jobs)

- Worker nominates a real person as guarantor.
- System generates a Guarantor Instrument (parties, scope, conditions, validity).
- Guarantor applies an electronic signature.
- Instrument status: Draft → Pending Signature → Active → Triggered / Expired / Revoked.
- Instrument can be queried and is linked to the worker/job.

5. Portable Work Credentials

- Workers can export their verified history and trust record.
- Workers can import previously verified credentials when joining a new platform.
- Practical credential-sharing mechanism (DID/VC optional later).

6. Escrow + Micro-Insurance (Platform-Independent jobs only)

- Funds placed in escrow via Vesti.
- Micro-premium paid from escrow.
- Cover activates only after premium is confirmed.
- Cover is voided if the protected flow is abandoned.

7. Job Completion Confirmation

- Both parties confirm completion → payment/outcome released.
- Customer timeout → auto-release.
- Active dispute → funds held and flagged for review.
- Reliable outcomes feed the Trust Scoring Engine.

8. Anti-Gamification Ratings (Platform-Independent)

- Reviews only possible after a completed, protected job.
- One review per job, tied to verified identity and job ID.
- Basic velocity and rate-limit protections.

9. Side-by-Side Demo Experience

- Same scenario shown On-Platform (with trust signals + guarantor) vs Off-Platform (no protection).

10. Identity (Leveraged)

- Integrate existing KYC providers (NIN, BVN, liveness, document).
- Store references, not raw identity data where possible.

P1 – Should Ship

- Worker dashboard / portable record view
- Basic individual profile + shareable verification link
- Simple platform configuration of baselines (API or lightweight UI)
- Clear audit trail of key actions

P2 – Out of Scope

- Escrow on platform-integrated jobs
- Institutional/platform-as-guarantor model
- Live insurance or CAC integrations
- Full production DID/VC
- Machine-learning models (rules-based formula only)

5. Key User Journeys

Platform Onboarding Validation

1. Platform sends worker data (normalised via API).
2. API returns decision (bool) + confidence.
3. Platform applies its own business rules using the result.

Personal Guarantor Creation

1. Worker nominates guarantor.
2. Instrument generated.
3. Guarantor electronically signs.
4. Status becomes Active and is queryable.

Platform-Independent Job

1. Parties agree on job.
2. Personal guarantor instrument created and signed.
3. Funds + micro-premium go into escrow (Vesti).
4. Cover activates after premium confirmation.
5. Job completed via dual confirmation.
6. Outcome recorded → trust score updates.
7. Rating window opens (anti-gamification rules applied).

Portable Credential

1. Worker exports record from Platform A.
2. Worker imports into Platform B during onboarding.
3. Platform B receives verified history instead of zero.
4.

6. Tech Stack

Layer	Technology
Frontend	Next.js + React + Tailwind
Backend	FastAPI (Python)
Database	Neon (PostgreSQL)
Escrow	Vesti (independent jobs only)
API docs:	Auto-generate from Zod/OpenAPI (or write a clean OpenAPI file)
Hosting	Backend: Render Frontend: Vercel
Identity	Existing KYC providers

7. Trust Output Contract (API)

JSON

```
{
 "decision": true,
 "confidence": 0.82,
 "band": "High",
 "status": "Building History" | "Scored",
 "signals_used": ["completion_rate", "dispute_count", ...],
 "explanation_summary": "Strong completion history with low disputes"
}
```

8. Success Criteria for Hackathon

- Live demo of side-by-side experience
- Working API that returns boolean + confidence
- End-to-end personal guarantor flow with electronic signature
- Portable credential import/export visible
- Platform-independent job with escrow (Vesti) + cover activation
- Job completion confirmation updating trust score
- Clear narrative: why staying inside the protected system is rational

FILE STRUCTURE:

```
/apps
  /api          ← Fastify/NestJS + Prisma + scoring engine
  /web          ← Next.js demo + dashboards
/packages
  /shared       ← Zod schemas, types, scoring formula
/docker-compose.yml
```

Trust Scoring Engine

- Pure TypeScript (or Python) functions
- Weighted formula + rules (no ML model required)
- Store raw outcomes → compute band on read or on write
- Keep the formula in one clear module so it can later be replaced or enhanced by the ML engine

Service	Approach for MVP
KYC	Mock server or sandbox of a real provider (e.g. simple endpoint that returns success + reference ID)
Insurance	Mock – just flip a `coverStatus` flag after premium is marked paid
Electronic Signature	Simple “I agree” + timestamp + recorded IP/user agent (store as evidence). No need for full DocuSign-style integration
PDF Guarantee	Generate on the fly and store/link

Data & Signal Strategy Across Platform Types

Vertical	Example High-Value Signals	Notes
Domestic	Completion, punctuality, safety incidents, repeat hire, quality evidence	Safety & reliability weighted highest
Handyman/Artisan	Job completion, quality flags, dispute rate, rework, on-time	Quality evidence critical
Ride-hailing	Trip completion, cancellation, rating, complaints	High volume, lower individual stakes
Logistics	Delivery success, damage reports, time windows	Similar to ride-hailing with asset risk

How these three fit together

Use Case	Primary User	Interface	Main Value	MVP Priority
Platform API	Gig / domestic platforms	API	Cutout reduction + risk control	Highest
Portable Reputation	Workers	Worker dashboard + credentials	Cross-platform reputation	High
Individual / Off-platform	Individuals & direct clients	Web profile + shareable link	Basic trust for informal hiring	Medium (seed + demo)